Alok Sen

Robotics & Control Engineer

RESUME_1401

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Total Experience: 1 year

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 1 year of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Simulink, PID Control, MATLAB, LiDAR, ROS 2, Motion Planning, Robot Kinematics, C++, Python, Gazebo, SLAM, PLC

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Robotics & Control Engineer | HelioNex Labs | Jul 2025 - Present

- Lowered navigation recovery events by 31% using improved obstacle handling
- Reduced robotic path completion time by 40% through route and controller tuning
- Validated 247 robot simulation scenarios with recorded sensor logs

PROJECTS

Motor Control Testbed | Gazebo, Simulink, LiDAR, Python

- Tuned PID loops and safety stops for repeatable positioning.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Warehouse Mapping Trial | MATLAB, Motion Planning, PID Control, LiDAR

- Benchmarked LiDAR SLAM configurations under moving obstacles.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Crescent Valley Technical University, 2025
Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

ROS 2 Navigation Workshop (2025)